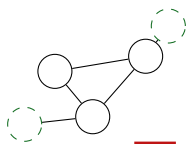
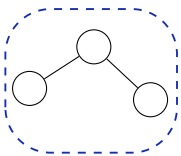


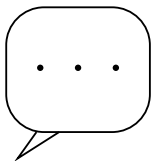
1. Adversary Observation Surfaces



1A. Protected graph G'

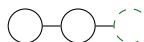


1B. Retrieved paths / subgraphs

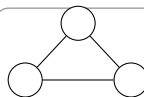


1C. Retrieval-grounded relation-QA output

2. Evaluated Attack Families



2A. Fixed-edge original-label inference



2B. Structure-only relation inference



2C. Sensitivity-policy flag inference



2D. Embedding-based relation inference

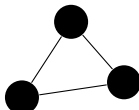


2E. Retrieval-grounded E2E relation exposure

3. Evaluated Privacy Risks



3A. Original sensitive relation-label recovery



3B. Structure-supported relation inference

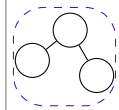


3C. Downstream relation exposure

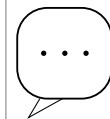
4. Evaluation Protocol



4A. Graph-level attacks on G'



4B. Path / subgraph evaluation



4C. Retrieval-grounded relation-QA evaluation



4D. Fixed splits, seeds and audit artifacts

Legend



Original node



Pseudonymized node



PRIVATE_RELATION
(protected label)

Key

G' = protected graph after TAFP

■ = observed / accessible by adversary

○ = pseudonymized / masked element